

Random variables and PMFs

A random variable turns each outcome into a number. Its PMF gives the chance of each number.

THE QUESTION

Once every outcome gives a number, such as the total of two dice, how likely is each number?

THE INTUITION

A random variable is a rule that gives every outcome one number. Several outcomes can give the same number, so $P(X = x)$ adds the chances of all outcomes that map to x . Listed over all values, those chances are the PMF.

KEY TAKEAWAY

$p_X(x) = P(X = x)$, the total chance of the outcomes with $X(\omega) = x$. A PMF is never negative and sums to 1.

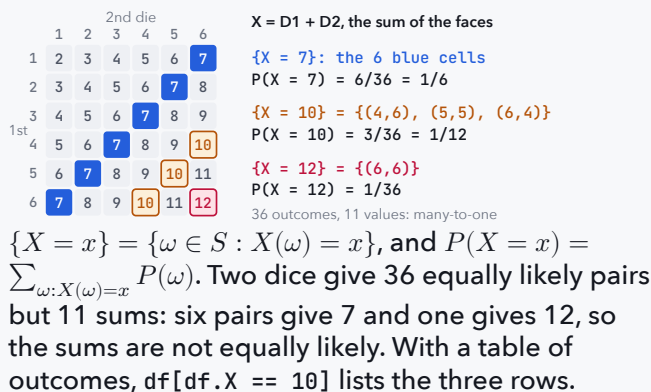
outcome ω → $X(\omega)$: a number → group equal values → PMF $p(x)$ → $Y = g(X)$: regroup → simulate: histogram $\approx$ PMF

1 Each outcome gets one number

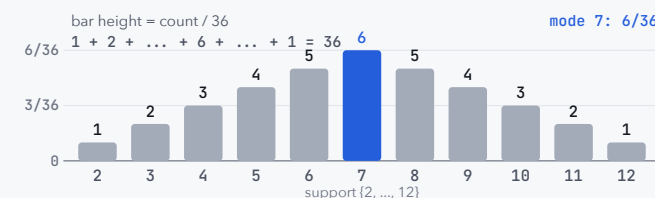


Flip a coin twice; X counts the heads. X is a function from outcomes to numbers, $X : S \rightarrow \mathbb{R}$: $\text{HH} \mapsto 2$, $\text{HT} \mapsto 1$, $\text{TH} \mapsto 1$, $\text{TT} \mapsto 0$. The outcome is random; the rule is fixed. Capital X names the variable, small x a value. Grouping by value gives $\{0: [\text{TT}], 1: [\text{HT}, \text{TH}], 2: [\text{HH}]\}$, so $P(X = 1) = 2/4$. An event's indicator (1 if it happens, else 0) is a random variable too.

2 $\{X = x\}$ is an event, so it has a probability

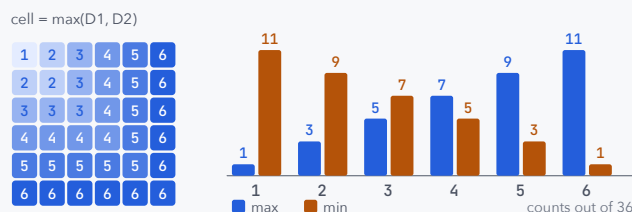


3 The PMF lists the chance of every value



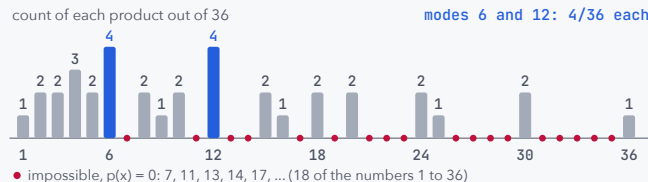
The probability mass function is $p_X(x) = P(X = x)$. Every PMF has $p_X(x) \geq 0$ and $\sum_x p_X(x) = 1$. Its support is the set of values with $p_X(x) > 0$, here $\{2, \dots, 12\}$. For a set of values A , add the bars: $P(X \in A) = \sum_{x \in A} p_X(x)$, so $P(X \geq 10) = (3 + 2 + 1)/36 = 1/6$. The most likely value, 7, is the mode.

4 Max and min: the odd numbers 1, 3, ..., 11



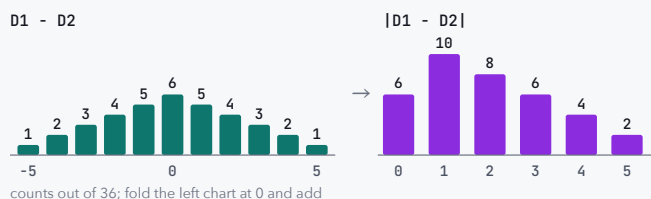
The max is k when both dice are at most k but not both at most $k - 1$: $(k^2 - (k - 1)^2)/36 = (2k - 1)/36$. So 6 is the likeliest max (11/36, 30.6%) and 1 the rarest (1/36). The min mirrors it, $P(\min = k) = (13 - 2k)/36$. The 36 outcomes are the same as for the sum; a new function gives a new PMF.

5 The product takes 18 values, with gaps



$X = D_1 D_2$ takes only 18 of the numbers 1 to 36. Values such as 7, 11, 13 and 32 are not in the support, so $p_X(7) = 0$. The modes are 6 and 12, with 4/36 each: $6 = 1 \cdot 6 = 6 \cdot 1 = 2 \cdot 3 = 3 \cdot 2$. Only one pair gives $25 = 5 \cdot 5$. The product is odd only when both dice are odd, $9/36 = 1/4$.

6 The difference, and its size $|D_1 - D_2|$

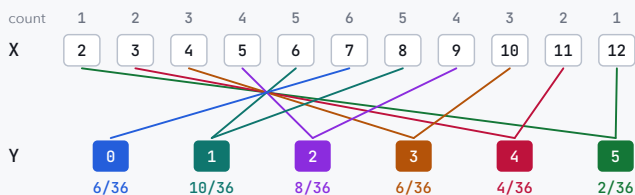


$D_1 - D_2$ runs from -5 to 5 with the triangle of the sum, shifted: $7 - D_2$ is again a fair die, so $D_1 - D_2 + 7$ has the PMF of the sum. Its size $|D_1 - D_2|$ folds the chart at 0 and adds the two sides: $p(0) = 6/36$ and $p(k) = 2(6 - k)/36$ for $k = 1$ to 5 . A gap of 1 (10/36) is likelier than a double (6/36).

New variables from old, and checking a PMF

A function of a random variable is a random variable. Its PMF comes from regrouping, and simulation checks it.

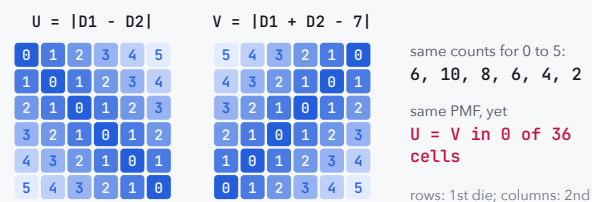
7 $Y = g(X)$: add the $p_X(x)$ that land on each y



If $Y = g(X)$, each outcome gets the number $g(X(\omega))$, so Y is a random variable too. Its PMF adds the chances of the x values that map to y : $p_Y(y) = \sum_{x: g(x)=y} p_X(x)$. With X the sum and $Y = |X - 7|$: $p_Y(1) = p_X(6) + p_X(8) = 10/36$. A one-to-one g , such as $Y = 2X + 1$, merges nothing: the bars keep their heights and move.

```
from itertools import product
from collections import Counter
from fractions import Fraction
S = product(range(1, 7), repeat=2)
cnt = Counter(abs(a + b - 7))
for a, b in S:
    {y: Fraction(c, 36)
     for y, c in sorted(cnt.items())}
# 0: 1/6, 1: 5/18, 2: 2/9, ..., 5: 1/18
```

8 Same PMF, different random variables



$U = |D_1 - D_2|$ and $V = |D_1 + D_2 - 7|$ have identical PMFs, yet $U = V$ on none of the 36 outcomes: (1, 1) gives $U = 0$, $V = 5$. (Equality would need $(2D_1 - 7)(7 - 2D_2) = 0$.) Likewise the two dice share one PMF but agree only on doubles, $P(D_1 = D_2) = 1/6$. A PMF says how often each value comes up, not which outcome gives it; such variables are "equal in distribution".

11 Pause and think

Is $p(1) = 0.5$, $p(2) = 0.3$, $p(3) = 0.3$ a PMF? No: the values add to 1.1.

Ten rolls of a die show no 3. Is $p(3) = 0$? No: a fair die misses 3 in ten rolls with $(5/6)^{10} \approx 16\%$.

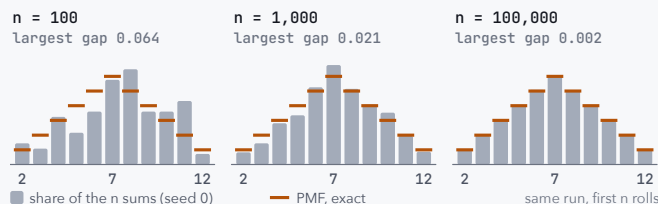
Two dice: which is likelier, max = 6 or sum = 7? Max = 6: 11/36 against 6/36.

Two dice: the most likely min? 1, with 11/36: the max chart read backwards.

X is $-1, 0$ or 1 , each with $1/3$. PMF of X^2 ? $p(0) = 1/3$, $p(1) = 2/3$: -1 and 1 merge.

Does $2D_1$ have the PMF of $D_1 + D_2$? No: $2D_1$ takes only 2, 4, ..., 12, each with $1/6$.

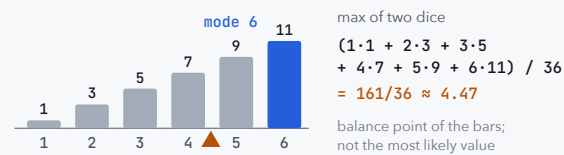
9 Simulated histograms approach the PMF



The PMF is exact. A histogram of n simulated values (each value's share of the n) estimates it and changes from run to run. In one run of two-dice sums, the largest gap to the PMF falls from 0.064 at $n = 100$ to 0.021 at 1,000 and 0.002 at 100,000. Short runs can miss possible values, so a zero bar in a histogram does not mean $p(x) = 0$.

```
rng = np.random.default_rng(0)
d = rng.integers(1, 7, (10_000, 2))
s = d.sum(axis=1)
vals, counts = np.unique(s, return_counts=True)
counts / len(s) # 7: 0.1697 (1/6)
pd.Series(s).value_counts(normalize=True).sort_index()
```

10 Densities, CDFs and means come later



X is discrete when its values can be listed, finite or countable: faces, counts, heads. A continuous X , such as a height or a waiting time, takes any value in an interval; each single value has $P(X = x) = 0$, and a density replaces the PMF (the continuous distributions sheet). Named PMFs (Bernoulli, binomial, geometric, Poisson) are on the next sheet; running totals $P(X \leq x)$ on the CDF sheet. A preview of the expectation sheet: the mean is the weighted average $\sum_x x p_X(x)$, 7 for the sum and 4.47 for the max.

X maps outcomes to numbers · $\{X = x\}$ is an event · $p(x) = P(X = x) \geq 0$, summing to 1 · $Y = g(X)$: add the $p(x)$ with $g(x) = y$ · histograms approach the PMF

Source: the course notebook "Random variables" (heads in two flips; sum, product and maximum of two dice; a fair die's PMF against histograms of simulated rolls), nipunbatra.github.io/psdv-teaching. The other examples and all numbers are our own, computed by the sheet's figure script; simulations use seed 0. Next: the discrete distributions sheet.